# HK IMO Training 2022-2023
# Problem Set 6 Suggested Solutions

**Problem 21** (Balkan MO 2016 Problem 3)**.** Find all monic polynomials $f$ with integer coefficients such that $p$ divides $2(f(p)!)+1$ for all sufficiently large primes $p$ for which $f(p)$ is a positive integer.

**Solution.** If $f(p) > p$ for some sufficiently large prime $p$, then $2(f(p)!)$ is a multiple of $p$, and hence $p \nmid 2(f(p)!)+1$, contradiction. In particular, this shows $\deg f \leqslant 1$. It is impossible that $\deg f = 0$ since no prime $p > 3$ divides $2(1!)+1 = 3$. Thus, we may assume $f(x) = x - c$ where $c \geqslant 0$ (as we need $f(p) > p$).

We need $p \mid 2((p-c)!)+1$ for all large prime $p$. By Wilson's theorem, we have

$$2((p-c)!)+1 = 2 \cdot \frac{(p-1)!}{(p-1)(p-2)\cdots(p-c+1)} + 1 \equiv 2 \cdot \frac{-1}{(-1)(-2)\cdots(-(c-1))} + 1 \equiv \frac{2(-1)^c}{(c-1)!} + 1 \pmod{p}.$$

This is $\equiv 0 \pmod{p}$ iff $p \mid 2(-1)^c + (c-1)!$. As $p$ can be arbitrarily large, this holds iff $2(-1)^c + (c-1)! = 0$. Thus, we need $2 \nmid c$ and $(c-1)! = 2$. This gives the only solution $c = 3$, which means $f(x) = x - 3$. $\square$

**Problem 22** (Canada MO 2014 Problem 2)**.** Let $m$ and $n$ be odd positive integers. We colour each cell of an $m \times n$ table black or white. Let $x$ be the number of rows with more black cells than white cells, and let $y$ be the number of columns with more white cells than black cells. Find the largest possible value of $x + y$.

**Solution.** If $m = 1$ (or $n = 1$), the answer is $n$. This can be attained if we colour all cells white, as $x = 0$ and $y = n$. Clearly, this is the largest possible case as we cannot have $x = 1$ and $y = n$ simultaneously.

If $m, n \geqslant 3$, the answer is $m + n - 2$. This can be attained if we colour all cells in the top left rectangle of size $\frac{m-1}{2} \times \frac{n+1}{2}$ black, colour all cells in the bottom right rectangle of size $\frac{m-1}{2} \times \frac{n+1}{2}$ black, and colour all other cells white. Then all $m-1$ rows except the middle one have more black cells, and all $n-1$ columns except the middle one have more white cells. This gives $x + y = m + n - 2$.

We now show that it is impossible for $x + y \geqslant m + n - 1$. If $x + y \geqslant m + n - 1$, WLOG assume $x = m$ and $y \geqslant n - 1$. Then there are at least $\frac{m(n+1)}{2}$ black cells and at least $\frac{(n-1)(m+1)}{2}$ white cells. As there are $mn$ cells in total, we have

$$mn \geqslant \frac{m(n+1)}{2} + \frac{(n-1)(m+1)}{2} = mn + \frac{n-1}{2} > mn.$$

This is a contradiction. $\square$

**Problem 23** (All-Russian MO 2015 Grade 11 Problem 6)**.** Let $a, b, c, d$ be real numbers such that $|a|, |b|, |c|, |d| > 1$ and

$$a + b + c + d + abc + abd + acd + bcd = 0.$$

Prove that

$$\frac{1}{a-1} + \frac{1}{b-1} + \frac{1}{c-1} + \frac{1}{d-1} > 0.$$

**Solution.** The condition can be rewritten as

$$(a+1)(b+1)(c+1)(d+1) = (a-1)(b-1)(c-1)(d-1).$$

Let $\alpha = \frac{a+1}{a-1}$, $\beta = \frac{b+1}{b-1}$, $\gamma = \frac{c+1}{c-1}$ and $\delta = \frac{d+1}{d-1}$. Note that $\alpha, \beta, \gamma, \delta$ are positive as $|a|, |b|, |c|, |d| > 1$. The condition is $\alpha\beta\gamma\delta = 1$. It is easy to obtain $a = \frac{\alpha+1}{\alpha-1}$, and so $\frac{1}{a-1} = \frac{\alpha-1}{2}$. So we need to prove

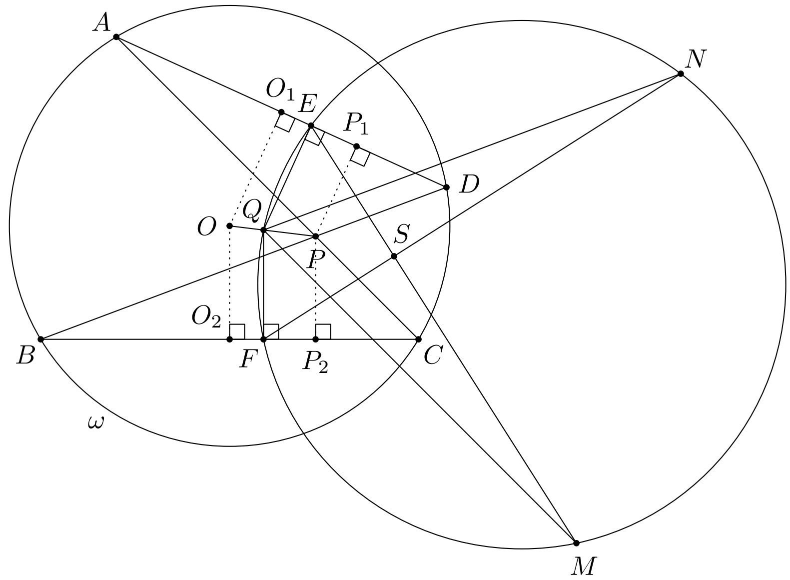
$$\frac{\alpha-1}{2} + \frac{\beta-1}{2} + \frac{\gamma-1}{2} + \frac{\delta-1}{2} > 0,$$

i.e. $\alpha+\beta+\gamma+\delta>4$. Indeed, by the AM-GM inequality, we have

$$\alpha+\beta+\gamma+\delta \geqslant 4\sqrt[4]{\alpha\beta\gamma\delta}=4.$$

Equality holds when $\alpha=\beta=\gamma=\delta=1$ (as $\alpha\beta\gamma\delta=1$). This is impossible as $\dfrac{a+1}{a-1}=1$ has no solution. Therefore, the inequality must be strict. $\square$

**Problem 24** (IGO 2022 Intermediate Problem 5)**.** Let $ABCD$ be a quadrilateral inscribed in a circle $\omega$ with centre $O$. Let $P$ be the intersection of two diagonals $AC$ and $BD$. Let $Q$ be a point lying on the segment $OP$. Let $E$ and $F$ be the orthogonal projections of $Q$ on the lines $AD$ and $BC$, respectively. The points $M$ and $N$ lie on the circumcircle of triangle $QEF$ such that $QM \parallel AC$ and $QN \parallel BD$. Prove that the two lines $ME$ and $NF$ meet on the perpendicular bisector of segment $CD$.




**Solution.** We assume $Q$ can be any point on the line $OP$ (not necessarily on the segment $OP$).

Let $S$ be the intersection of $ME$ and $NF$. We have

$$\angle SEF=\angle MEF=\angle MQF=90^\circ-\angle ACB=\angle OAB.$$

By symmetry, $\angle SFE=\angle OBA$. Therefore, $\triangle SEF\sim\triangle OAB$.

Let $O_1, P_1$ be the projections of $O, P$ onto $AD$, and let $O_2, P_2$ be the projections of $O, P$ onto $BC$ respectively. Note that we have $\dfrac{AO_1}{O_1D}=1=\dfrac{BO_2}{O_2C}$, $\dfrac{AP_1}{P_1D}=\dfrac{BP_2}{P_2C}$ (as $\triangle PAD\sim\triangle PBC$) and $\dfrac{O_1E}{EP_1}=\dfrac{OQ}{QP}=\dfrac{O_2F}{FP_2}$. Combining these, we have $\dfrac{AE}{ED}=\dfrac{BF}{FC}$. Let $T$ be the centre of spiral similarity mapping $AD$ to $BC$ (which is a fixed point). This spiral similarity also maps $E$ to $F$ (as $AE:ED=BF:FC$). Therefore, there exists a spiral similarity with centre $T$ mapping $AB$ to $EF$. This spiral similarity maps $O$ to $S$ as $\triangle OAB\sim\triangle SEF$. As the locus of $E$ is a straight line, the locus of $S$ is also a straight line.

Suppose $Q$ is the point such that $E=A$. Then $F=B$ as $AE:ED=BF:FC$. Since $\triangle SEF\sim\triangle OAB$, we must have $S=O$, which lies on the perpendicular bisector of $CD$.

Suppose $Q$ is the point such that $E=D$. Then $F=C$ as above. Since $\triangle SEF\sim\triangle OAB$, we have $SE=SF$, i.e. $SD=SC$. So $S$ lies on the perpendicular bisector of $CD$ (and $S\neq O$ in this case).

Therefore, the locus of $S$ is the perpendicular bisector of $CD$. $\square$